

Does Random Option Ordering Change Language Model Answers? A Study of Permutation Invariance in Multiple-Choice QA

Anonymous ACL submission

Abstract

Permutation sensitivity, the tendency of a language model’s answer to shift when the positions of options in a multiple-choice prompt are reordered, threatens the reliability of model evaluation. Existing evidence documents this instability but offers no mechanistic account of when or why it occurs. This paper proposes a systematic study of permutation invariance that will characterize the conditions under which option order affects predictions. The planned evaluation will compare candidate explanations across a set of models and benchmark tasks, treating option-order variants as the controlled intervention. The central question is whether sensitivity reflects a shallow positional bias or a deeper sensitivity to contextual distribution. Without such an account, reported multiple-choice accuracies may conflate model competence with artifacts of prompt construction, and the proposed experiments are designed to resolve this ambiguity.

1 Introduction

Large language models are increasingly used for multiple-choice question answering, yet their responses may depend on the order in which answer options are presented. A model that reliably selects the correct answer under one ordering could fail under another, even when the semantic content is unchanged. This option-order sensitivity is consequential because it undermines the trustworthiness of models deployed in high-stakes settings, where users expect a consistent answer regardless of surface-level formatting. Moreover, such sensitivity complicates evaluation: benchmark scores that are computed under a single canonical ordering may overstate a model’s true competence. The concrete problem addressed in this study is therefore to determine the extent to which random permutations of answer options alter model predictions, and to measure the consistency of model behavior across these orderings, as illustrated in Figure 1.

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Figure 1: Large language models give different multiple-choice answers when option order changes; the figure poses the study’s evaluation question of measuring prediction consistency across random permutations of answer choices.

Despite the practical importance of option-order sensitivity, existing work does not settle whether it reflects a systematic flaw or incidental variation (). Prior studies of prompt sensitivity have largely focused on instruction wording or example selection, leaving the specific effect of answer-option permutation underexplored (). Analyses that do examine ordering often treat it as a nuisance variable to be controlled rather than as an object of study in its own right. Consequently, there is no established account of how prevalent order-induced answer changes are across models, nor of whether certain models or question types are disproportionately affected. Without such evidence, it remains unclear whether current evaluation practices, which typically fix a single ordering, produce stable estimates of model competence or conceal systematic inconsistencies that a different ordering would reveal.

To address this gap, we propose a systematic study of permutation invariance in multiple-choice question answering. Our approach evaluates whether models produce consistent answers when the order of answer options is randomly permuted, treating the full set of permutations as the object of analysis rather than a nuisance factor. We define permutation invariance operationally as the stability of a model’s selected answer across repeated orderings of the same question. The con-

tributions of this study are threefold. First, we introduce a measurement protocol for quantifying order-induced answer variation. Second, we apply this protocol across a range of models to characterize the prevalence and magnitude of sensitivity. Third, we examine whether specific model families or question types exhibit differential susceptibility, providing a basis for understanding the underlying causes and for designing more robust evaluation practices.

The central evaluation question of this study is whether permutation invariance holds for large language models on multiple-choice question answering, and if not, how severe the violations are. Specifically, we ask: does randomly reordering answer options change a model’s selected answer, and if so, how often and under what conditions? The claim boundary is deliberately narrow. We do not assert that permutation sensitivity, if observed, stems from any particular mechanism, such as positional bias, token-level confounds, or training artifacts. Instead, the study is designed to establish the empirical facts: the prevalence of answer changes across permutations, the consistency of model behavior under repeated orderings, and the extent to which sensitivity varies across models and question types. Within this boundary, the study aims to provide a reproducible measurement of permutation invariance that can inform evaluation design, without overclaiming causal explanations.

2 Related Work

The closest methodological thread concerns how large language models respond to superficial variations in their input. Prior work has examined interventions on input structure, including the ordering of answer options in multiple-choice tasks (), and some studies report that models can produce different outputs when the same options are presented in a different sequence (). This thread primarily documents the phenomenon and proposes remedies such as prompt engineering and fine-tuning strategies (). However, it does not systematically characterize when the effect is large, when it vanishes, or which aspects of the model’s decision process it might reflect. The boundary of this thread is therefore descriptive rather than mechanistic, leaving open the question of whether option-order sensitivity is a stable property of the model’s reasoning or a byproduct of its decoding procedure. This open question motivates the controlled study pro-

posed in this paper, which will vary only the option permutation while holding all other inputs fixed.

A separate thread addresses the evaluation of multiple-choice question answering (), where the standard protocol samples either a single option ordering or a small fixed set of orderings for each item. These benchmarks () typically treat option position as a nuisance variable, averaging accuracy across repeated presentations or reporting a single pass without controlling for permutation. As a result, the evaluation literature does not provide a direct measure of how much a model’s answer depends on the specific sequence in which options appear, nor does it establish whether observed score differences across orderings exceed the noise of the evaluation procedure itself. The distinction for this paper is that it will treat option ordering as the independent variable rather than a controlled nuisance, measuring the full distribution of outcomes across all permutations of the same item while holding every other input component fixed.

Together, these two threads define the space this paper occupies. The methodological thread () establishes that option-order sensitivity is a real and observable phenomenon but stops short of explaining when or why it occurs. The evaluation thread (), by contrast, assumes the effect is small enough to ignore and therefore does not measure it directly. This paper sits at the intersection: it takes the phenomenon documented by the first thread as its object of study and applies the controlled measurement discipline advocated by the second. Specifically, it will isolate option permutation as the sole variable, examine the full distribution of outcomes across all orderings of a fixed item, and report the resulting sensitivity without relying on aggregate scores that could conceal it. This position is neither a new mitigation technique nor a new benchmark, but a systematic characterization of a previously under-measured property of multiple-choice answer generation.

3 Method

We investigate whether language models exhibit permutation invariance under random reorderings of answer options in multiple-choice question answering. Our approach compares model outputs across systematically shuffled option orders to determine whether predictions remain stable or shift with the presentation sequence. We operationalize the research problem through a controlled evalua-

tion protocol that isolates the effect of option ordering while holding all other input characteristics constant. This overview establishes the experimental design used throughout the remainder of the section, defining the core procedure that subsequent paragraphs specify in detail.

For each item in the evaluation set, we obtain a base prediction by presenting the multiple-choice question with a reference option ordering. We then generate a set of permuted variants by applying distinct random permutations to the answer options, producing a candidate pool of reordered prompts. For each variant, the model’s class score $s_k(x)$ is recorded for every option k , where x denotes the input prompt after permutation. We define permutation invariance as the consistency of the model’s argmax choice across all orderings of the same question. The procedure fixes the prompt template, decoding configuration, and scoring function, while the tested variable is exclusively the option ordering.

We fix the number of random permutations applied to each question and the random seed used to generate them, ensuring that every item receives the same set of reorderings. The prompt template, including the question stem, option labels, and instruction phrasing, remains constant across all trials. The tested variable is exclusively the ordering of the answer options. We record the model’s selected option and its associated class score for each permuted prompt, storing these outputs for downstream analysis. This design separates the fixed components of the evaluation from the manipulated variable, allowing any observed differences in predictions to be attributed to option ordering rather than to incidental variation in the input format.

We hypothesize that language models are not inherently invariant to option ordering, and that shuffled permutations can alter predictions in a systematic rather than random manner. One plausible mechanism is that the model attends disproportionately to option positions and surface cues, such as the label letter or the relative placement of the correct answer, rather than relying solely on semantic content. Under this account, the class score $s_k(x)$ assigned to a given option may depend not only on its content but also on its position within the prompt, so that reordering changes the relative ranking of options even when the underlying question is unchanged. We treat this mechanism as a hypothesis to be tested rather than an established property of the models under study.

4 Experiments

We will evaluate permutation invariance on multiple-choice question answering benchmarks. For each question, we will construct all possible option orderings via exhaustive permutation. The primary metric will be the fraction of questions where the predicted answer remains unchanged across all orderings; the secondary metric will measure the variance of class scores. Baselines will include a random-order strategy and a fixed canonical ordering. We will use greedy decoding with a fixed seed and will report results per dataset and model size.

The main comparison will contrast permutation-invariance rates across the benchmark datasets and model sizes described above. Table 1 will present the primary metric for each condition, reporting the fraction of questions with unchanged predictions alongside the secondary variance measure. Figure 2 will display the corresponding distributional trend across option-order count. For every cell, the unavailable value is denoted as xx. No measurements are currently available, and we will compute these values once the experimental pipeline is executed.

No experiment results are available in the hosted project, so this paragraph cannot report any observed pattern. The evidence currently establishes the evaluation protocol alone: the primary metric, the secondary variance measure, the exhaustive permutation procedure, and the planned baselines. The evidence does not establish whether option ordering affects predictions, whether any effect varies by dataset or model size, or whether the secondary metric behaves consistently with the primary one. These questions remain open until the planned comparison is executed and the measurements are supplied.

Robustness checks will focus on two dimensions. First, we will vary decoding temperature to test whether permutation-invariance rates remain stable beyond the greedy setting. Second, we will repeat the analysis using a fixed canonical ordering as the reference condition to isolate ordering effects. No significance thresholds or confidence intervals have been specified, and no effect-size measurement is available; all corresponding values are denoted as xx.

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Figure 2: Planned primary result analysis showing the distributional trend of permutation-invariance rates across option-order count, with unavailable values denoted as xx; no measurements are yet available.

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Table 1: Planned main experimental comparison across methods and evaluation criteria.

5 Discussion

Table 1 is designed to compare permutation-invariance rates across option-order counts and evaluation criteria, while Figure 2 is designed to display the distributional trend of these rates across the same dimension. Together, these artifacts test whether sensitivity to option ordering varies systematically with the number of alternatives presented. Because no measurements are yet available, the current evidence gap is explicit: we cannot yet determine whether the hypothesized trend exists, nor whether its direction, if any, is monotonic. We therefore treat the possibility of a systematic ordering effect as a hypothesis only, to be confirmed or rejected once the planned experiments are executed and the placeholders are populated.

A primary limitation is that the planned study measures permutation invariance only through aggregate rates across option-order counts, which may obscure item-level variation in sensitivity. The design does not yet isolate whether observed instability stems from genuine order effects or from model variance across repeated samplings, since the evaluation protocol has not fixed a decoding temperature or a number of stochastic passes. A further threat to validity is that the candidate pool generated by reordering options is not equivalent to the set of examples admitted to a refit, so conclusions about selection behavior must remain scoped to the former. Finally, without held-out answer distributions we cannot distinguish a model that

flips between plausible alternatives from one that degrades to near-random choice.

6 Conclusion

We presented a controlled study of permutation invariance in multiple-choice question answering, defining the core comparison, the samples, and the candidate baselines that the planned evaluation will test. The design distinguishes the fixed implementation choices from the variables under study, and the expected sensitivity mechanism is stated explicitly as a hypothesis rather than an observed outcome. The principal limitation is the absence of uploaded experimental results: no quantitative evidence currently supports a claim about how often answer distributions change under random option reordering or which baseline best predicts such changes. Future work should run the specified evaluation, report the supported comparisons, and extend the analysis to other model families and prompt formats.